COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE312 | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 13/08/2026 | **DATE SUBMITTED:** | 13/08/2026 |
| **NAME:** | Abon, Benedict Aldous<br>Garcia, Ron Allen<br>Tiong, Martin Andrei | **INSTRUCTOR:** | Engr. Robin Valenzuela |

## PRELIM EXAM

### 1. PROCEDURE OUTPUT

**Problem: Voice-Based Gender Classification using Machine Learning Algorithms**

1. Create your own dataset using different voices from different genders* saying "Hello".
   Optional 1: You may include audio from videos scraped from the web.
   Optional 2: If you can find public datasets, add that to your own.
2. Train classification models and test on new data from people not in your dataset.
3. Analyze the results and provide recommendations.
4. You may use published research papers as reference for your work; all references must be cited.

## I. Introduction

Human speech is not just considered as a tool of communication and expression, but also a rich and unique identifier for every individual. The human ear represents a natural voice sensor that is capable of distinguishing people and their gender. As technology advances, speech recognition has been evident in various applications like gender classification in human to machine interaction like phone calls, video categorization with tagging, and registration systems, which works by extracting human voices and evaluating its metrics like pitch, frequency, and other acoustic properties (Livieris et al., 2019). As the gender recognition approach using Long Short-Term Memory (LSTM) already exists, a lighter but accurate model is needed for easier applicability and modification (Ertam, 2019). Consequently, this project addresses the need for a specialized machine learning model optimized for extracting voice qualities and establishing an accurate classification model which classifies male and female voices.

## II. Methodology

The methodology includes the step-by-step process involved in creating the machine learning model of the project. It covers the sequential process and different stages of the model development process, from data gathering, data cleaning, to the model training and evaluation. Each process is done in an orderly manner to ensure that the data is properly prepared and usable for the model to be trained properly, this helps in making sure that the model is reliable and accurate to improve the effectiveness of the model. It also mentions the tools and resources used in accomplishing the project, along with its general function and specific contribution in the project.

### A. Step-by-Step Process

#### a. Data Acquisition

The project mainly utilizes YouTube videos containing movie clips of male and female movie characters saying the word ‘hello.’ These parts are extracted as one to three second audio clips through a Python script. This script includes the YoutubeDL library that downloads Youtube videos locally by adding in video links, the Pydub library that extracted the audio from the linked video, and the Faster Whisper library that sliced each extracted audio to the point where the word ‘hello’ is heard.

The average pitch is used as the primary metric for the classification model, which is why it is extracted in each audio file. The librosa library is used to extract this specific variable, because it is capable of reading analog signals like sound and converting them into digital signals in the form of acoustic metrics. This metric is then stored in a Pandas DataFrame for easier manipulation and usage in model training.

**b. Data Cleaning**

The created voice properties DataFrame is formalized and cleaned. The gender column is also created to categorize these metrics, which is essential in the next step for identifying the gender distribution in the training and testing data. Missing and duplicated values were also removed from the dataset to avoid inconsistencies.

**c. Model Training**

After cleaning the dataset, model training was made. The logistic regression algorithm will be used for the project. This algorithm is a simple and effective method used in creating binary classification models. With the dataset having 200 records, a 70/30 split is used to distinguish the training and testing data for the project.

**d. Model Evaluation**

The model evaluation involves validating the model by assessing the model accuracy, classification report, logistic regression graph, and confusion matrix. To further test the model, it also involved a Python script that takes in new voice data and predicts the speaker's gender and returns its average pitch.

## B. Tools and Resources

**a. Python**

Python is a general purpose programming language famous for its simplicity, ease of use, readability, and versatility, relying on english keywords to make it more accessible for beginners. It was used in the project as the core programming environment for the audio-based gender recognition pipeline. Notable libraries used are:

- **Librosa**: It was used as the core library for reading the .wav files and converting said files into the specific sample rate. It was also used to run the *pyin* algorithm to extract the pitch across speech frames
- **Scikit-learn**: It was used to split the dataset into 70% training and 30% testing while maintaining class balance. It was also utilized as the binary classifier that maps the pitch between the 2 genders(male and female).
- **Matplotlib and Seaborn**: handled the plot layouts, layouts for the figures, and overall visualizations for the confusion matrix.
- **Glob:** A python module used to parse the file paths and retrieve all .wav file links as a bulk from their directories.
- **Pandas**: Handled the dataframe structures, extraction of pitches, and labels for cleaning, filtering and model input.
- **Numpy**: Used for array computations and handling missing pitch frames.

**b. YouTube**

YouTube is a global video sharing and social media platform, mainly used by users to watch and post content. The platform was used in this study to extract the 'hello' audios needed from a video from said platform.

**c. Google Colab**

Colab is a free, cloud-based Jupyter notebook environment provided by Google. It was used in the study to perform the extraction of the audio needed, from the raw video fetched from Youtube.

d. **Google Drive**
Drive is a storage space for different files, photos, and videos provided by Google. It was utilized in the study as a form of storage for the arranged audio files based on their gender.

e. **Gemini AI**
Gemini AI is a generative artificial intelligence model and a virtual assistant chatbot developed by Google. The AI tool was used as a form of assistance in achieving the objectives of the project. The AI was used to process the extraction of the needed data.

## III. Results

The following points further explains the importance of methods used in evaluating the model performance. It also presents the results gathered from these methods and briefly tackles them one by one.

### A. Model Accuracy

The Logistic Regression model is used in the project. This model is defined as a supervised machine learning algorithm that is used to predict what category each prediction falls under. This makes it applicable in defining categories between predictions (Spiceworks, 2022). In application, the model was evaluated on a test set of samples to assess its ability to classify gender based solely on the average fundamental pitch of their voices, represented by 'mean_pitch_hz' in the dataset. As we are dealing with male and female voices, using the logistic regression model was appropriate as it involves a binary classification model, where we predict if a voice is made by a male or female speaker.
After running the model training process, the model is able to achieve an accuracy rate of 83%.

```
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')

Accuracy: 0.83
```

Figure 3.1: Model accuracy result

### B. Classification Results

The classification results present the precision, recall, f1-score, and support of the model. The precision answers the percentage of correct predictions in the model. The recall shows how many positive cases were correctly identified by the model. The f1-score shows how many positive predictions were correctly identified by the model. The support returns the number of actual occurrences of the class in the dataset. In short, these values represent the main classification metrics on a per-class basis, giving deeper intuition of the classifier behavior over global accuracy. In this project, we used this to further assess how correctly the model predicted each category in the dataset.
As seen in the classification metrics, the model shows a balanced performance across the different classes. The female category achieved a precision of 0.85, a recall of 0.82, and an F1-score of 0.84. For the male category on the other hand, the model achieved a precision of 0.81, recall of 0.85, and a resulting F1-score of 0.83.

| | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Female | 0.85 | 0.82 | 0.84 | 28 |
| Male | 0.81 | 0.85 | 0.83 | 26 |
| Accuracy | | | 0.83 | 54 |
| Macro average | 0.83 | 0.83 | 0.83 | 54 |
| Weighted average | 0.83 | 0.83 | 0.83 | 54 |

Figure 3.2: Classification report table, showing the model's precision, recall, f1-score, and support

### C. Logistic Regression Graph

The visualization of the logistic regression is used in the project mainly showing the probabilities between category 0 and 1 in a format humans can understand. It can also help us identify any outliers in the predictions made by the model. In this scenario, we used the logistic regression graph to find out to which mean pitch values do male and female voices fall under, and it showed a clear separation between these voices in terms of their pitch.

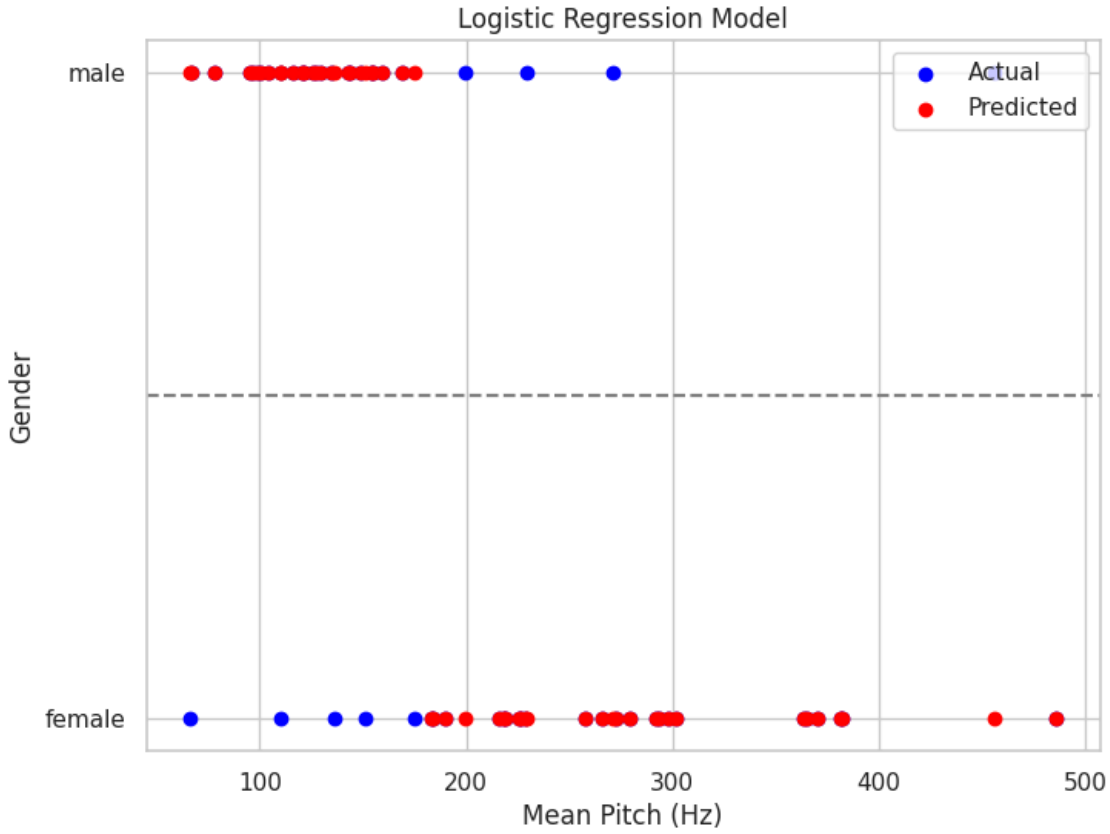



Figure 3.3: Logistic regression graph to visualize gender classification

### D. Confusion Matrix

The confusion matrix is mainly used for assessing the performance of the model per class, where we understand which classes are most easily identified and confused by the model.

The confusion matrix revealed a total of 9 misclassifications out of the 54 tests. 23 female audios were correctly identified as female and 22 male audios were correctly identified as male. For the misclassifications, 5 female samples were classified wrongly as male, while 4 male samples were misclassified as female. Errors can be mostly attributed to the frequency of the voice. A standard logistic regression model fits a single linear decision boundary. Higher pitched male voices and lower pitched female voices in between are prone to this misclassification.

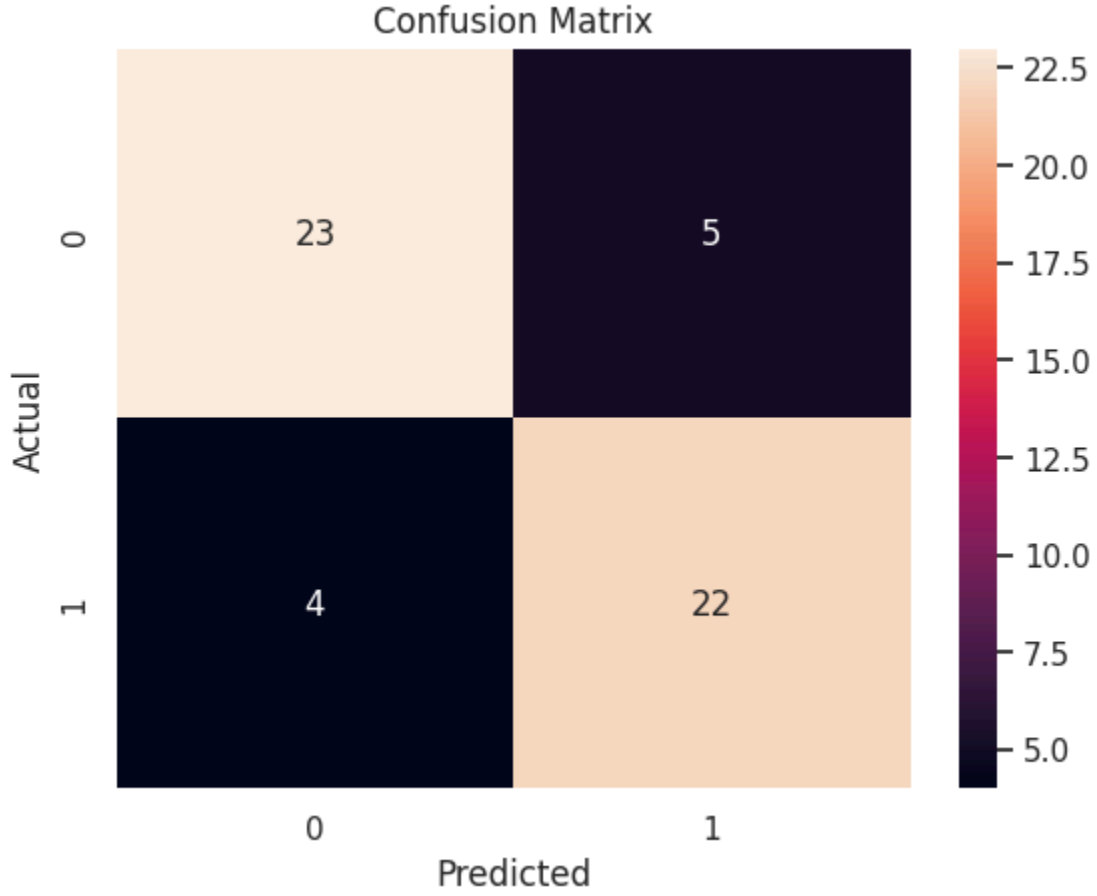



Figure 3.4: Confusion matrix, where 0 represents female voice and 1 represents male voice

## IV. Discussion

The model evaluation results show that the model has a strong distinction of gender through their voices. As female voices naturally have higher pitch than of men, we can say that the average vocal pitch is a great indication of gender. This finding was also evident in the logistic regression graph, where male voices mainly sit in the range of 0-200 hz mean pitch, while female voices sit in the range of 200-500hz mean pitch. However, some misclassifications were present in the model. Based on the confusion matrix, 5 female voice samples and 4 male voice samples were misclassified. With this, we can assess that using voice frequency as a sole variable for classifying gender could lead to some errors, since some male people may have high-pitched voices and some female people may have low-pitched voices as well.

The study may be interpreted as a great starting point for creating a gender classification model with voices. To improve its accuracy and performance, it is suggested to improve the model through adding more variables that will strongly assist in classifying gender through voice. This will also involve revisions on how the model will be manipulated. A larger dataset containing more audio files from different sources can be introduced in order to create a more accurate model. Revising the scripts in scraping audio files and making their signals clearer can also improve the data quality used in the project.

In summary, the project successfully addresses the problem of gender classification by creating a specialized machine learning model. We are able to create a gender classification model using voices and establish accurate predictions on male and female voices by acquiring and calculating the mean pitch of their voices. This created the fundamental framework for the development of lighter and faster voice recognition machine learning models for human to computer interaction medium.

## V. Appendix

### A. References


Ertam, F. (2019). An effective gender recognition approach using voice data via deeper LSTM networks. Applied Acoustics, 156, 351–358. https://doi.org/10.1016/j.apacoust.2019.07.033

Livieris, I. E., Pintelas, E., & Pintelas, P. (2019). Gender Recognition by Voice Using an Improved Self-Labeled Algorithm. Machine Learning and Knowledge Extraction, 1(1), 492–503. https://doi.org/10.3390/make1010030

Nandita and B. Jain, "A Comprehensive Review of Machine Learning Techniques for Voice-Based Gender Recognition," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-7, doi: 10.1109/ICCCNT56998.2023.10306939.

Spiceworks. Everything You Need to Know About Logistic Regression. (2022, April 8). Spiceworks Inc; Spiceworks. https://www.spiceworks.com/soft-tech/what-is-logistic-regression/


### B. Member Contributions

| Member Name | Contribution |
|---|---|
| Abon, Benedict Aldous | ● Model Training Process<br>● Documentation |
| Garcia, Ron Allen | ● Data Extraction<br>● Documentation |
| Tiong, Martin Andrei | ● Data Extraction<br>● Data Transformation<br>● Documentation |

### C. Raw Data Links

The following links show the YouTube link of videos, where the study extracted the required audio files for the project.

People saying HELLO in movies PART2: https://www.youtube.com/watch?v=DVrhPUWPnZI

People saying HELLO in movies PART 3: https://www.youtube.com/watch?v=8iU1iqIrqC0

**D. Dataset Links**

The following links redirects to the datasets used in the model training process of the project. These datasets are stored in a Google Drive folder.
Male Voices Dataset:
https://drive.google.com/drive/folders/1mG39VDNN6GldauUjUTQL1rM0Kot8V7N9?usp=sharing
Female Voices Dataset:
https://drive.google.com/drive/folders/1H0LavO-yHjqEKCCAA8hvAvhhL3kP548l?usp=sharing

**E. Colab Links**

The following links are the Colab Notebooks used for the project.
Data Acquisition Python Notebook
CPE312_Group 1_Prelim Exam_Data Gathering.ipynb
Data Cleaning and Model Training Python Notebook
CPE312_Group 1_Prelim Exam_Model Training.ipynb

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** Gemini, Quillbot

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** Web-scraping for needed data, Grammar correction, Code debugging

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** Gemini was used to assist in the extraction of the voices that we used in our dataset, extracting it as a wav file. The organization by gender was done by the researchers manually for it to be usable in training the model.

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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